1. General Information

- **Full Name / Student ID:** Nguyen Duc Minh / 2640105
- **Course:** Information Systems
- **Assignment:** Labwork 1
- **Environment:** PostgreSQL 18 on Windows Operating System

2. Implementation Steps and Commands

Step 1: Clone the repository from GitHub

- **Description:** Download the `test_db` database repository from GitHub.
- **Command used:**

```
git clone https://github.com/datacharmer/test_db
```

- **Output:**

```
Cloning into 'test_db'...
remote: Enumerating objects: 262, done.
remote: Counting objects: 100% (130/130), done.
remote: Compressing objects: 100% (48/48), done.
remote: Total 262 (delta 100), reused 82 (delta 82),
   pack-reused 132 (from 2)
Receiving objects: 100% (262/262), 73.47 MiB | 3.19
   MiB/s, done.
Resolving deltas: 100% (148/148), done.
Updating files: 100% (36/36), done.
```

Step 2: Load data into PostgreSQL Server

- **Description:** Execute the script to create the schema and load 300,000 employee records into PostgreSQL using the `postgres` account.
- **Command used:**

```
PGUSER=postgres bash load_employees_db.sh
```

- **Output:**

```
DROP DATABASE
CREATE DATABASE
You are now connected to database "employees" as user
   "postgres".
info
------------------------------
```

```
CREATING DATABASE STRUCTURE
(1 row)


psql:F:/file/University/Master/Information system/
   test_db/postgresql/employees.sql:22: NOTICE:  table
    "dept_emp" does not exist, skipping
psql:F:/file/University/Master/Information system/
   test_db/postgresql/employees.sql:22: NOTICE:  table
    "dept_manager" does not exist, skipping
psql:F:/file/University/Master/Information system/
   test_db/postgresql/employees.sql:22: NOTICE:  table
    "titles" does not exist, skipping
psql:F:/file/University/Master/Information system/
   test_db/postgresql/employees.sql:22: NOTICE:  table
    "salaries" does not exist, skipping
psql:F:/file/University/Master/Information system/
   test_db/postgresql/employees.sql:22: NOTICE:  table
    "employees" does not exist, skipping
psql:F:/file/University/Master/Information system/
   test_db/postgresql/employees.sql:22: NOTICE:  table
    "departments" does not exist, skipping
DROP TABLE
CREATE TABLE
CREATE TABLE
CREATE TABLE
CREATE TABLE
CREATE TABLE
CREATE TABLE
CREATE VIEW
CREATE VIEW
info
----------------
SCHEMA CREATED
(1 row)

Done loading employees database.
```

Step 3: Database Integrity Verification

- **Description:** Use the SHA-256 hash algorithm to verify whether the record counts and CRC checksums perfectly match the original dataset.
- **Command used:**

```
psql -U postgres -d employees < test_employees_sha2.
   sql
```

- **Output:**

```
You are now connected to database "employees" as user
   "postgres".
info
----------------------
TESTING INSTALLATION
(1 row)

CREATE EXTENSION
NOTICE:  table "expected_values" does not exist,
   skipping
NOTICE:  table "found_values" does not exist, skipping
DROP TABLE
CREATE TABLE
CREATE TABLE
INSERT 0 6
table_name  | expected_records |
                            expected_crc
--------------+------------------+----------------
employees    |           300024 | 21f5d0...
departments  |                9 | 377c5d...
dept_manager |               24 | 3a4e69...
dept_emp     |           331603 | 34548e...
titles       |           443308 | a9e940...
salaries     |          2844047 | 4e99e6...
(6 rows)

CREATE FUNCTION
INSERT 0 6
table_name  | found_records |
                             found_crc
--------------+---------------+-------------------
employees    |        300024 | 21f5d00...
departments  |             9 | 377c5d7...
dept_manager |            24 | 3a4e697...
dept_emp     |        331603 | 34548ee...
titles       |        443308 | a9e940e...
salaries     |       2844047 | 4e99e69...
(6 rows)

table_name  | records_match | crc_match
--------------+---------------+-----------
employees    | OK            | ok
departments  | OK            | ok
dept_manager | OK            | ok
dept_emp     | OK            | ok
titles       | OK            | ok
salaries     | OK            | ok
(6 rows)

summary | result
---------+--------
CRC     | OK
count   | OK
```

```
(2 rows)

DROP FUNCTION
DROP TABLE
```

3. Conclusion

- The `employees` database was successfully initialized and verified on the PostgreSQL database management system.
- All 6 tables (`employees`, `departments`, `dept_manager`, `dept_emp`, `titles`, `salaries`) completely matched in record counts and CRC values (`CRC Match: OK`, `Record Match: OK`).